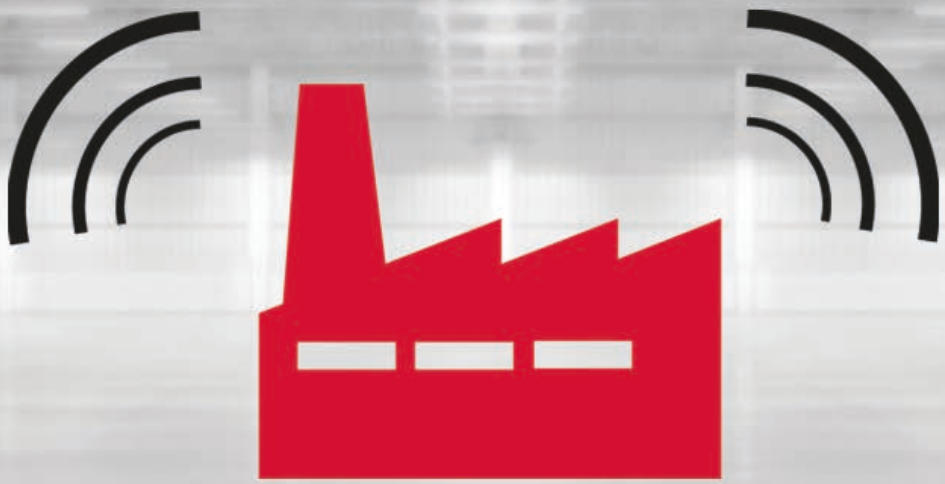
Industry 4.0



SMAHRT METROLOGY

IN THE FUTURE OF MEASURING

Metrology is getting smart. Smart metrology is equipped with local intelligence and networked.

This what EXACTLY means to us.

- 0 +

Mahr

EXACTLY

INDUSTRY 4.0

It‘s about the future and therefore about the further development of the concept of Industry 4.0 and the **Smart Factory** up to the **SMAHRT Metrology**, Mahr's answer to Industry 4.0 and thus to Metrology 4.0!

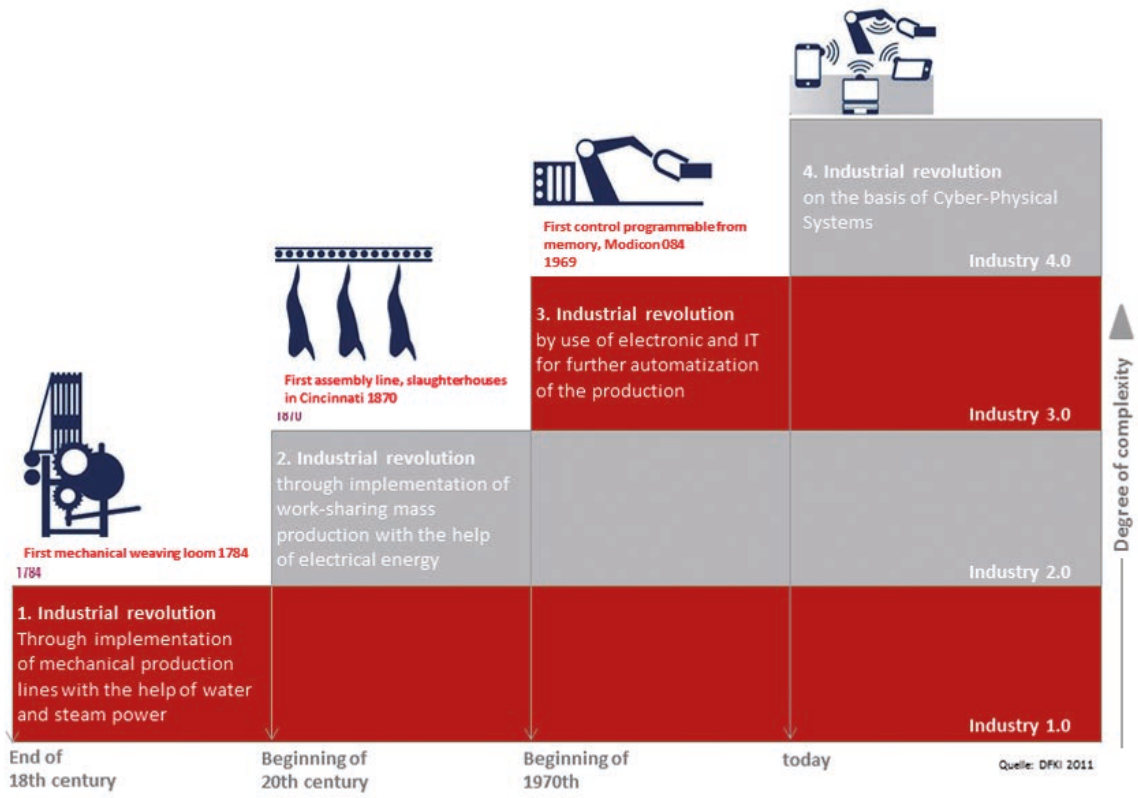


INDUSTRY 4.0

THE DEVELOPMENT TOWARD INDUSTRY 4.0

Digitization has changed our world. The internet and modern technology increasingly shape the manufacturing industry. We are facing a significant change – the threshold of Industry 4.0. The vision of Industry 4.0 shows that new requirements on production systems and machines are needed. They must be flexible to adapt with continuously changing manufactured products. That means: Production will be more individual, flexible and faster. This leads to many starting points for Metrology 4.0.

Metrology is getting **SMAHRT!** The production of Industry 4.0 builds upon smart metrology. Smart metrology is equipped and networked wit local intelligence.



Not only is smart measuring required, but automatic, fast and user-friendly measurement is also of great importance. Examples include:

- the wireless calipers **SMAHRT Gages**
- customer solutions **SMAHRT Solution** and
- new worldwide: the mobile surface measuring unit **MarSurf PS10**, a real **SMAHRT SURF PS10** roughness measuring unit.

SMAHRT with expanded local intelligence, all in the spirit of Industry 4.0 and Metrology 4.0.

www.mahr.com/Services/Production metrology/Know-how/Industry 4.0

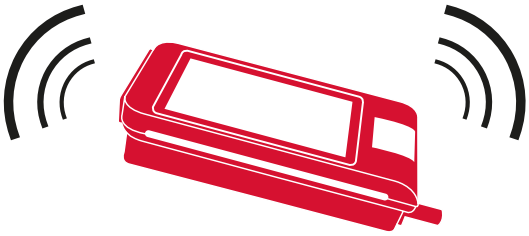
INDUSTRY 4.0

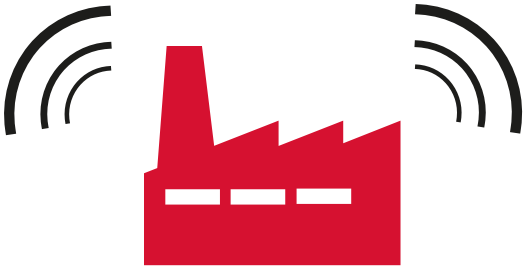
THE DEVELOPMENT TOWARD INDUSTRY 4.0

Industry 4.0 - the networked factory

The forth industrial revolution not only includes the development of the transportation belt in the factory, but changes the world outside of the factory gates as well. The entire logic of production is changing.

Intelligent machines and products, storage systems and resources are consequently interlinked by IT along the entire value chain, starting from logistics to production and marketing, ending with service.



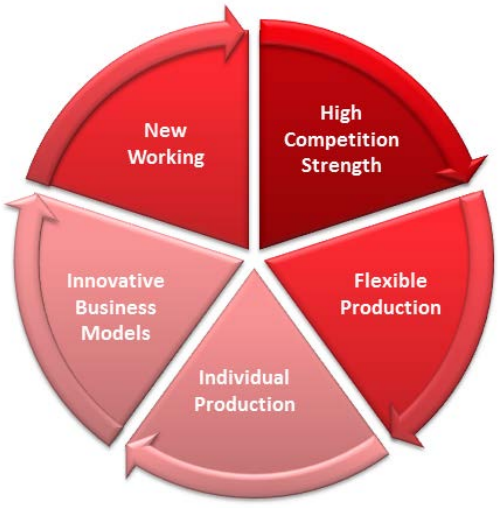


More individual, more flexible & faster

The first revolution came about with water power and the steam engine, giving people access to a wide range of goods for the first time.

Industry 4.0 will facilitate extensive individualization, radically changing how products will be manufactured and sold. Information is now a matter of course, accesible to everyone and no longer a privilege.

The vision of Industry 4.0 shows that new requirements on production systems and machines are needed. They must be flexible in terms of manufactured products, which are continuously changing. That means: Production will be more individual, flexible and faster.



Five arguments for Industry 4.0

INDUSTRY 4.0
SMART FACTORY

Industry 4.0 also means **Smart Factory**, which in turm leads to **SMAHRT Metrology** with:
SMAHRT solutions, **SMAHRT gages** and the **SMAHRT Surf PS10.**

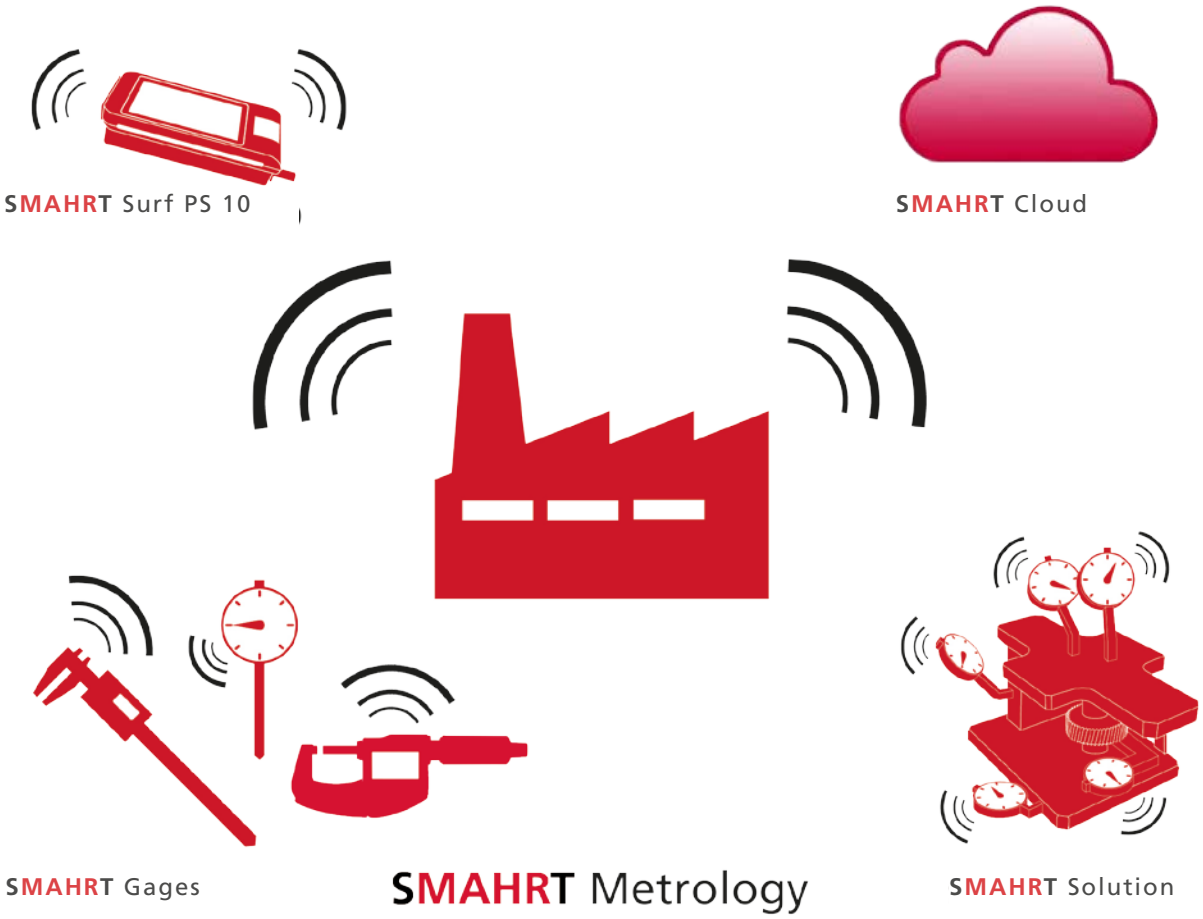
The term **Smart Factory** also comprises the concept of Condition Monitoring. This is based on a periodic or a permament detection of the state of the machine by measurement and analysis of physical quantities, e.g. vibations, temperatures, position / approximation.

Condition Monitoring generally has two objectives:

- **"Process Reliability"**
- **"Machine Efficiency"**

The mobile roughness measuring unit **MarSurf PS 10**, for example, records roughness profiles, shows the measuring result on a 4.3" touch, color dipslay and saves the measuring record in PDF format. The measuring data can be communicated via USB cable or wireless trasmission. Handling is as easy as a smartphone.

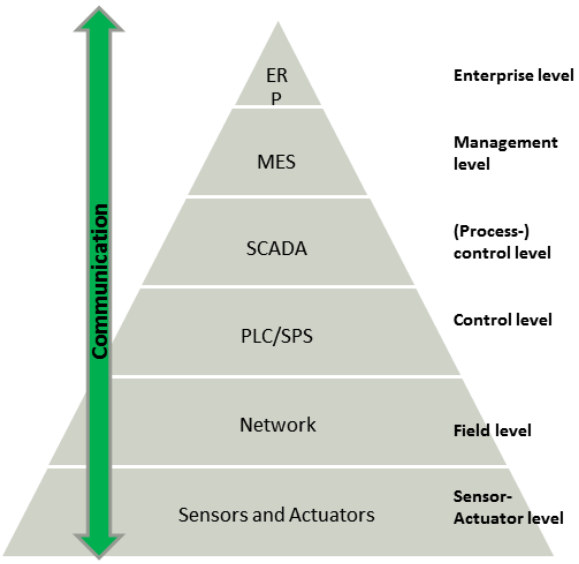
With the standardized Gear Data Exchange format (GDE format), SMAHRT cylinder wheel geometries and measurement data can be exchanged.



INDUSTRY 4.0
TARGETS AND REQUIREMENTS

Framework conditions
Communication leads to an increase in data within the production processes already at the lowest level.

Requierements for Industry 4.0
- Components must be able to provide (gather) the required data
- Seamless communication from the sensor to the internet
- Infrastructure in terms of transporting the data through all levels



ERP – Enterprise Resource Planning
ERP is business-management software that an organization can use to collect, store, manage and interpret data from many business activities, including product planning, cost, manufacturing or service delivery, marketing and sales, inventory management, shipping and payment. The aim is to ensure an efficient operational value creation process and a continuous optimized control of business and operational processes.

A core function of ERPS is to ensure that all materials and components needed for the manufacturing process will be at the right place at the right time and in the right quantity. Today this is only possible through the help of IT systems based on modern information and communication technology.

MES – Manufacturing Executing System
MES are used on a process-oriented operating level of a multi-dimensional production management system. MES differ from similar efficient systems like ERP in that they have direct connection to the distributed systems of the process automation and enables the management, steering, control or monitoring of production in real time. This includes classic data collections and formats such as production data acquisition (PDA), machine data acquisition (MDA) and personal data collection, but also all other processes that have an immediate impact on the manufacturing / production process.

SCADA – Supervisory Control and Data Acquisition
SCADA is a computer system for collecting and analyzing real-time data. Scada systems are network control systems that monitor, control and optimize industrial plants.

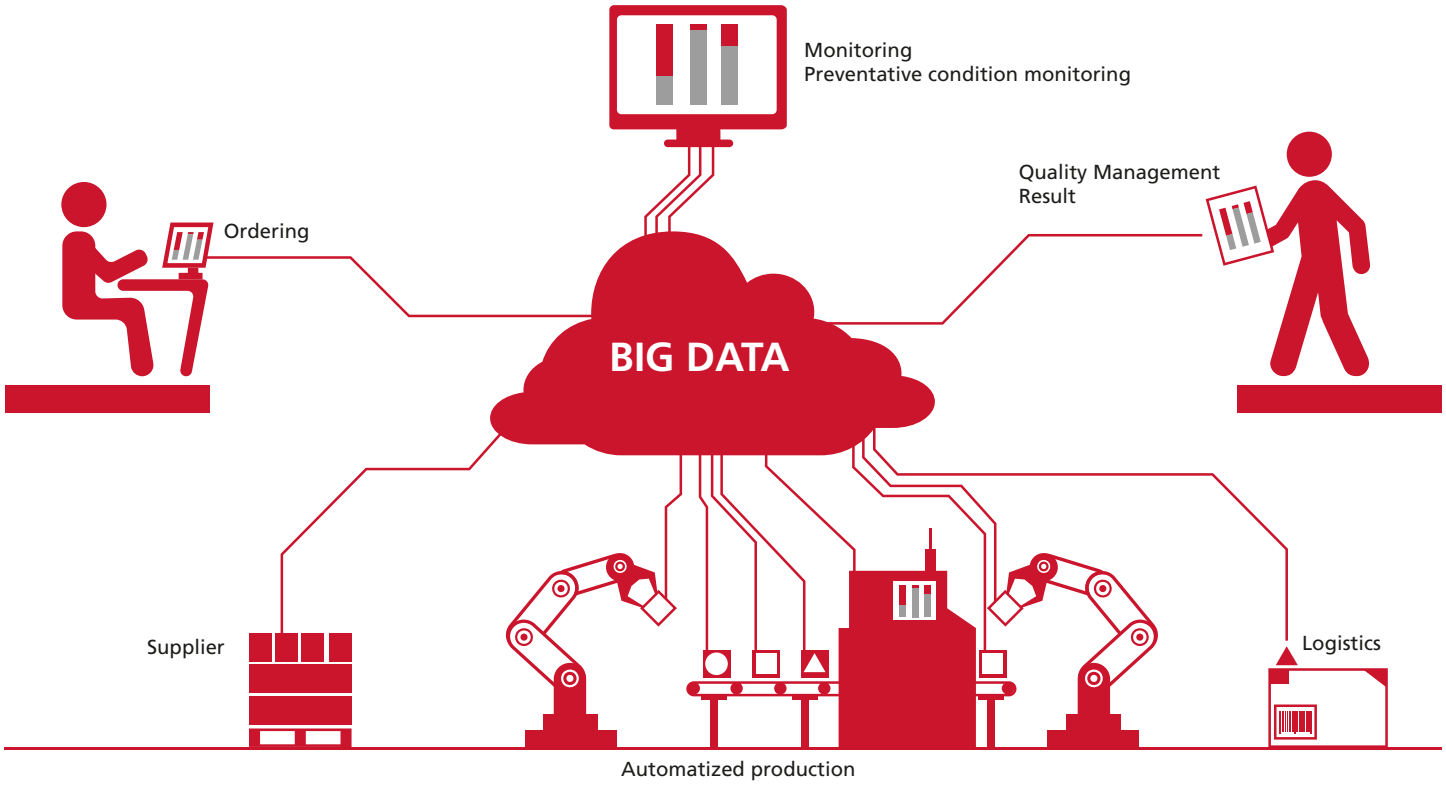
PLC – Programmable Logic Controller
A programmable logic controller (PLC) is a digital computer that is used for controlling or regulating a machine or system. For several years it has superseded the the “hardwired” control in most fields.

INDUSTRY 4.0
SMART FACTORY

Producing quality and decreasing overhead costs

The objective of all activities will be to produce quality and as a consequence to reduce overhead costs. This results in the follwing approaches for machines and instruments within the environment of dimensional metrology:

- Automation as a component that aims to reduce inspection costs
- Having measuring processes under control with the objective to reduce inspection costs
- Unlimited communication means networked systems respectively “Cloud Monitoring”
- A look into the machine: Local intelligence enables a preventive status monitoring
- Integrated Systems are networked units with sensors for self-monitoring



INDUSTRY 4.0

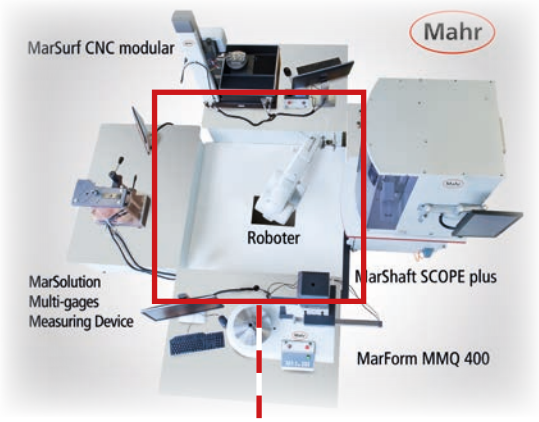
AUTOMATION - LOWERING TESTING COSTS

The measuring machines from Mahr can be connected with production machines. This enables autonomous adjustment of the production when there are variant tolerances. The assurance of workpiece quality, which is now free from operating errors, ensures a higher level of process reliability.

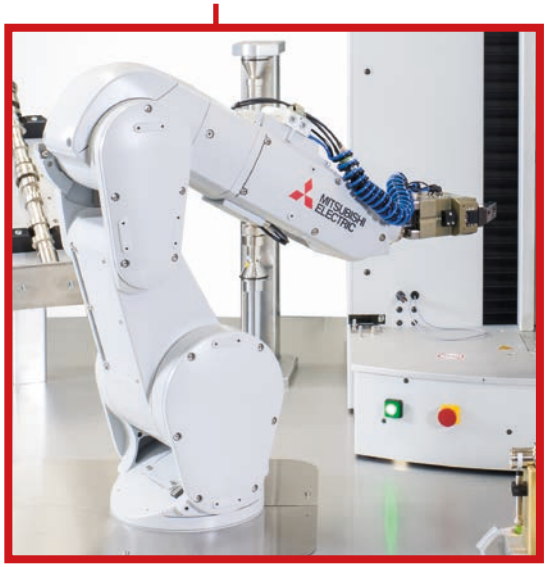
The vision pronounces a flexible, automated and connected machine in production. Mahr delivers the suitable answer for quality assurance: The measurement of camshafts on the shaft measuring system **Marshaft Scope 600 *plus* 3D**, including workpiece loading by robot is a perfect example of what measuring looks like within a production line.

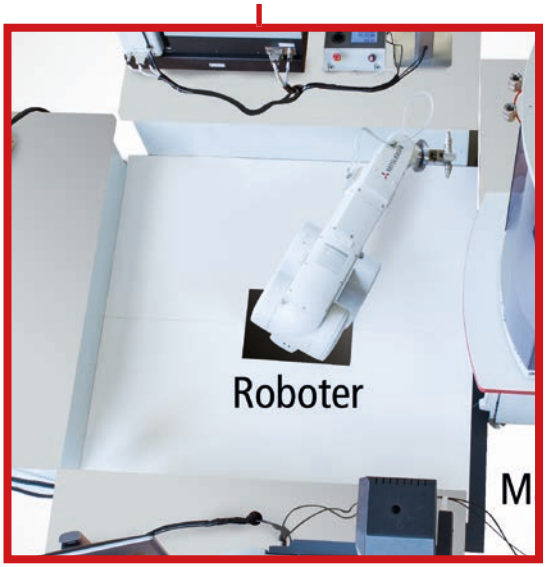
Mahr shows with its measuring station **Marsurf CNC *modular*** flexible and connected measuring of various workpieces: Automatic part recognition, part marking and robot handling are new possibilities of measuring technology in current and future production systems.





Loading and unloading by robot, integration of the measuring machine in the production line



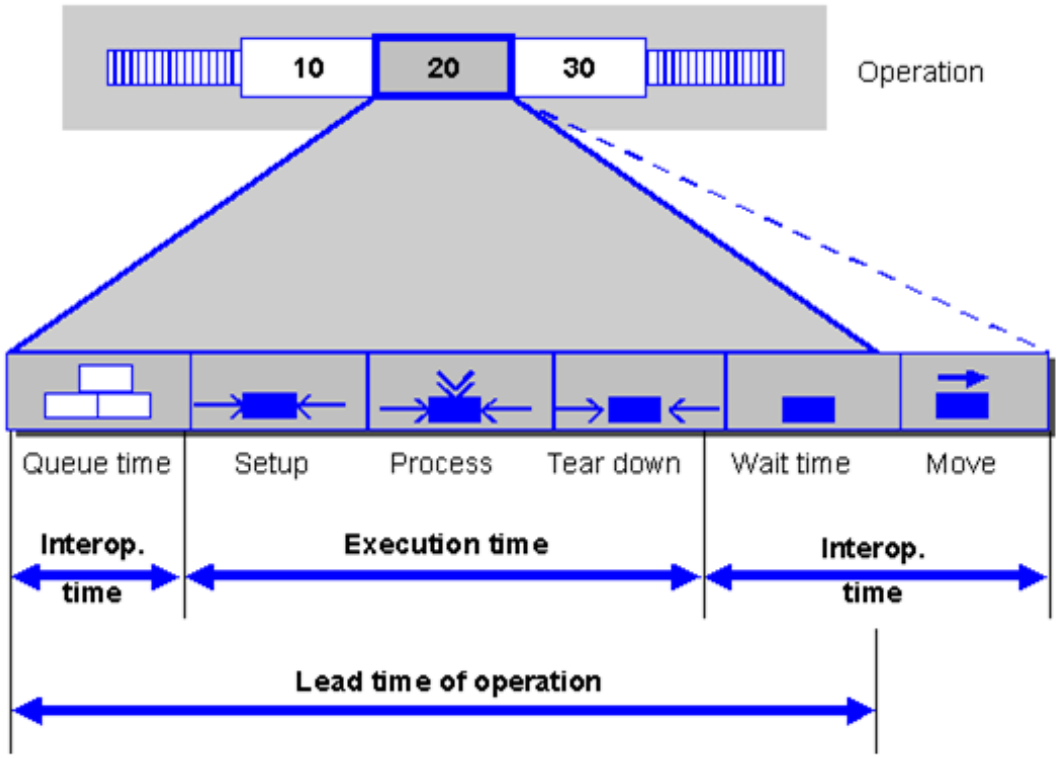


INDUSTRY 4.0

AUTOMATION - LOWERING TESTING COSTS

Today it is common practice for a plurality of measurement programs to be locally stored on a measuring machine that is intended for e.g. five different workpieces. These are “mutations“ and copies of the original programs and only the responsible worker knows exactly which measurement program is right for the machine.

The more unique the measuring program, the more accurate the measuring devices can be measured. Under this aspect, the future integration of the instrument into the organizational structure of the ERP system will take place. In addition, the integration is taken into account in the capacity of industry to ensure that only one measurement program is available on the measuring machine at the end of the integration process. This measurement program is then replaced by the new, to be executed measurement program.



Work planning in PPS system

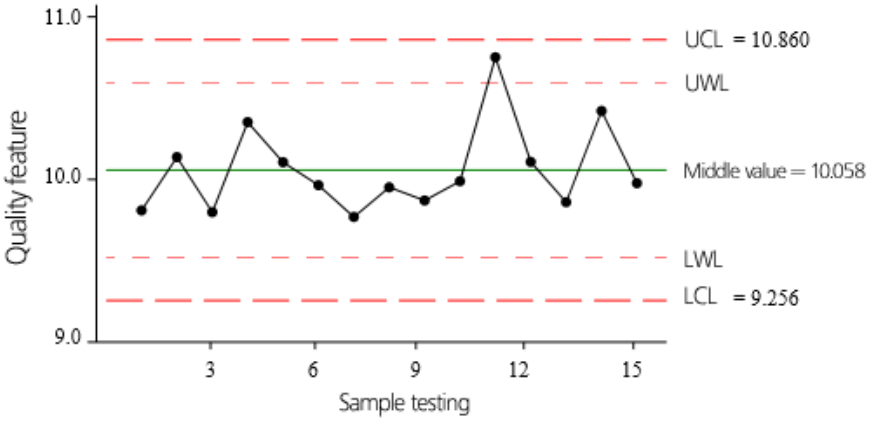
Organizational integration

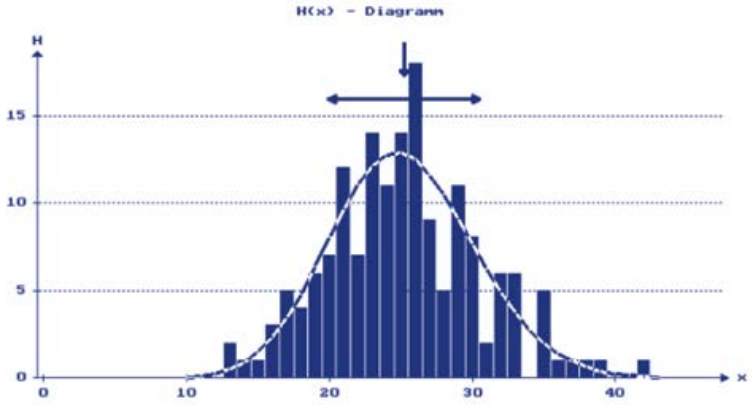
- Control and capacity management of the measuring technology by ERP/PPS systems
- Measuring program administration through ERP systems (unambiguity of measuring programs is relevant for quality!)

INDUSTRY 4.0

UNDERSTANDING AND MASTERING MEASURING PROCESSES

In this context, the Bosch booklet 10 is one of the most important guides to be mentioned. It includes among other things descriptions like the procedure of a %GRR method, which allows only a 1/40th of the tolerance as a variation between the repeated individual measured value. With a tolerance of 1 µm, this will be no more than 14 nanometers between two repeated measurements on the workpiece that has been removed and inserted again. These requirements result in the development of metrology to a higher incidence of multi-axis measuring machines, because only then can the operator influence be reduced to the minimum.





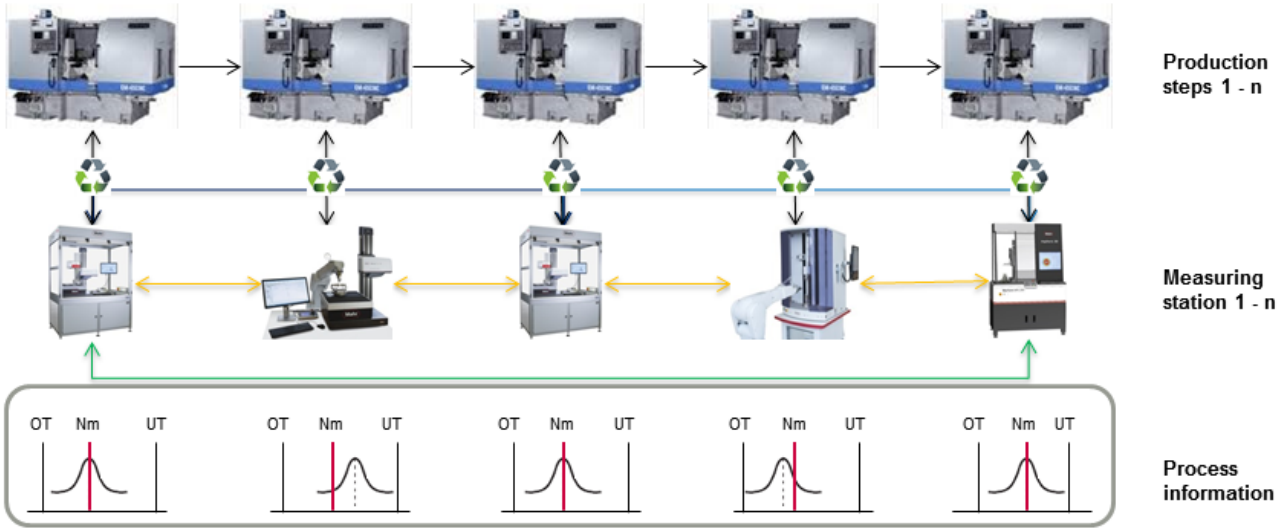
- Greater measuring reliability through automation with multi-axis measuring machine
- Higher measuring accuracy due to minimized operator influence
- Higher measuring accuracy to accelerate the transient process with SOT/SOP
- The basic error of the measurement decisively influences the closed loop!
- A successful closed loop depends on the quality of the measurement! (Process capability of the measuring technology)

INDUSTRY 4.0

UNDERSTANDING AND MASTERING MEASURING PROCESSES

Linking the measured values from different manufacturing steps

- Assessment of the process status (process capability of the measuring technology)
- Visualization in the control station for measuring tecnology
- Rules of the adaptive scope of testing and/or
- Testing tolerances for the next production steps wit the objecive: reduction of testing costs/overhead costs



The **Closed Loop**, meaning the control of the machine through the results of the measuring station is now already state of the art. By networking information, communication and thus an influence between the individual measuring stations will take place.

If the process is exactly or very close to the nominal size, the sampling frequency, i.e. the frequency of measurements, reduces the downstream inspection process or even allows a greater tolerance for the subsequent process.

If the process deviates from the ideal situation, it is reversed, i.e., it is switched back to the "normal testing process". This requires in addition to the required intelligence also the visualization of the processes of new process control stations, which also include the metrology processes to a greater extent.

In future, therefore, the closed loop will not only be limited to the combination of machine vs. measuring station, but also extended to communication between the stations and the PPS system.

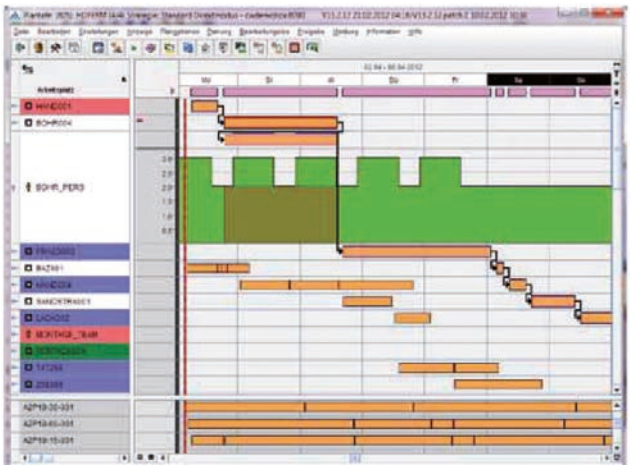
INDUSTRY 4.0
COMMUNICATION

Communicating without boundaries

This requires measuring devices that communicate with each other, that request data and send data, which then alter and influence the measurements themselves. Tolerance limits of prefabrication, for example, are adjusted depending on the tolerance utilization.

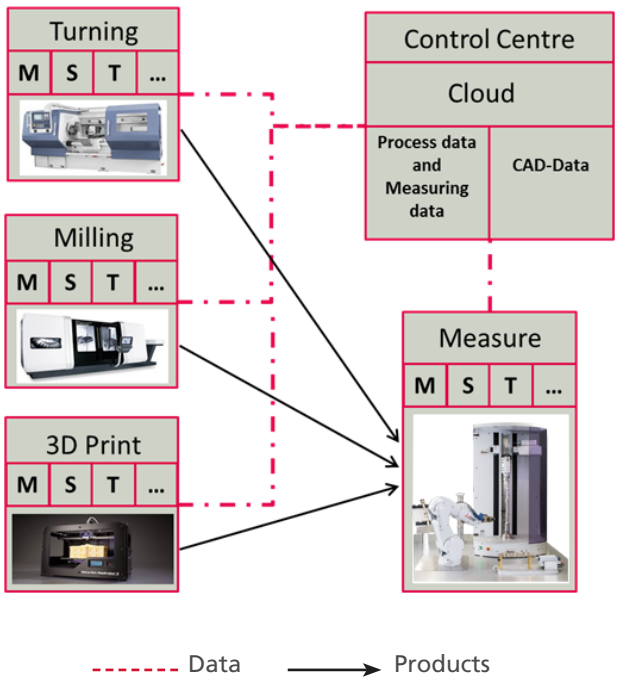
In the cloud, the visualization of the data is personalized and adapted to the current situation. Only as much information as is currently required and not as much as possible is displayed at the control station.





Process screening and process control station

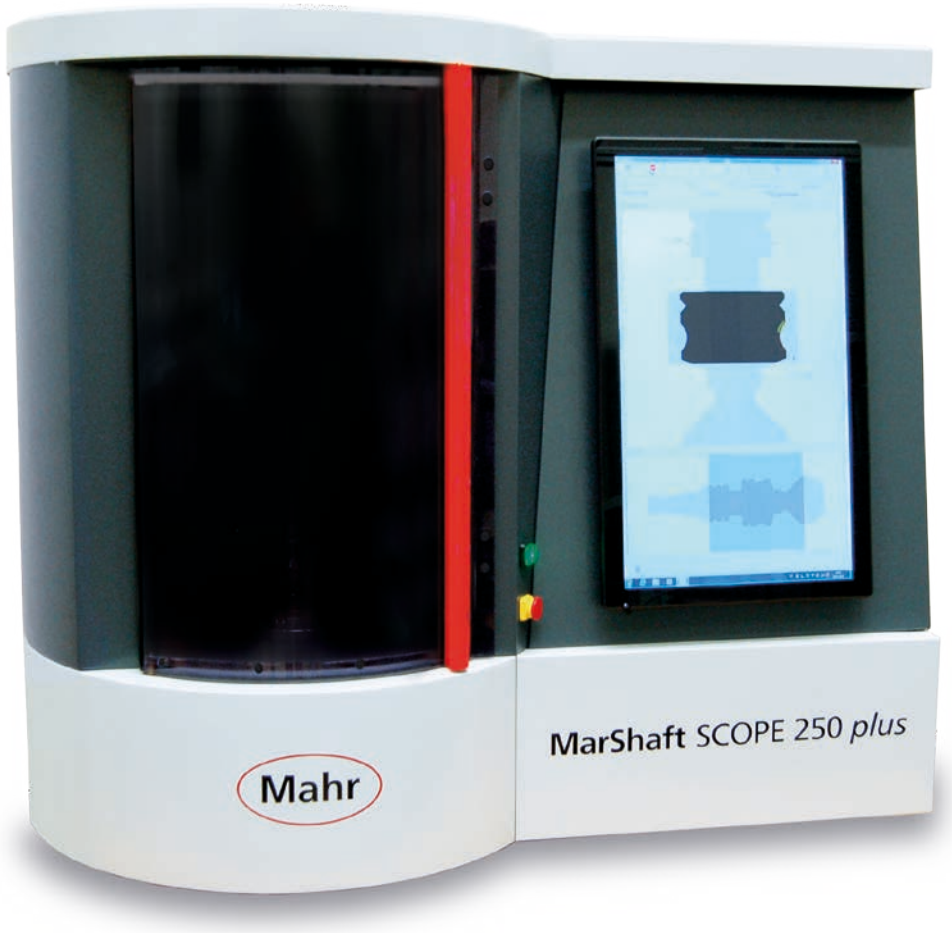
- 100% control or random sampling
- Control of crucial characteristics
- Tolerance specification for crucial characteristics
- Differentiate between real effects of random products - draw the correct conclusions from the data
- Process correction based on all gathered process data



INDUSTRY 4.0

OUTLOOK: PREVENTIVE CONDITION MONITORING

- Measuring axes are **SMAHRT**, have decentral intelligence and control themselves locally
- Mesasuring axes recognize changes comparatively
- Measuring machines know their componenents
- Measuring machines send information independently or on demand for just-in-time maintenance
- **Objective:** Minimization of maintenance and repair costs, increasing efficiency
- **Objective:** Optimization of ability
- **Objective**: Optimization of resources in the supplier's customer service
- **Objective**: Optimization of resources in the customer's factory maintenance



INDUSTRY 4.0

VISION OF INTEGRATED SYSTEMS

Industry 4.0 is based on the exchange of information

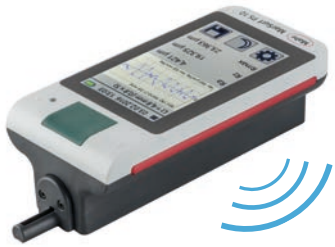
- Different components including various communication technologies become one networked unit.
- Machines possess additional sensors for self-monitoring, which are not necessarily needed (e.g. microphone chip to monitor warehouses, heat sensors on drives).
- Machines possess radio modules and chips to communicate.
- All components communicate among each other and share data.
- A "chipped" machine component is able to announce its upcoming outage and therefore initiate the production as well as the assembly of a replacement.

Producing quality

- Higher quality due to Industry 4.0
- Re-use and transparency of knowledge
- Real-time information and increased punctuality
- Faster and more efficient processes
- Increased capacity and shorter cycle times



INDUSTRY 4.0
SMAHRT METROLOGY



SMAHRT Surf PS 10

The **MarSurf PS 10** from Mahr is a compact roughness measuring instrument for mobile use. It is simple and intuitive to use and therefore very user-friendly. With the large, illuminated touch display, the small and handy measuring device can be used as easily as a smartphone. With 31 parameters, it offers the performance scope of a lab unit.

www.mahr.com
WebCode 21820



The new freedom in measuring: Cable-free measuring directly in production and wireless transfer of measuring data accelerates the quality assurance process. Mahr offers for this purpose wireless measuring devices as "**SMAHRT Products**" with local intelligence for the networked factory of the future. Wireless data transfer simplifies the acquisition and documentation of measuring data – also and especially in the networked factory of Industry 4.0. With the click of a button on the measuring device, keyboard, timer function, remote control or per footswitch the acquired data is sent from the measuring unit to the wireless receiver and is processed.

www.mahr.com
WebCode 20068



SMAHRT Gages

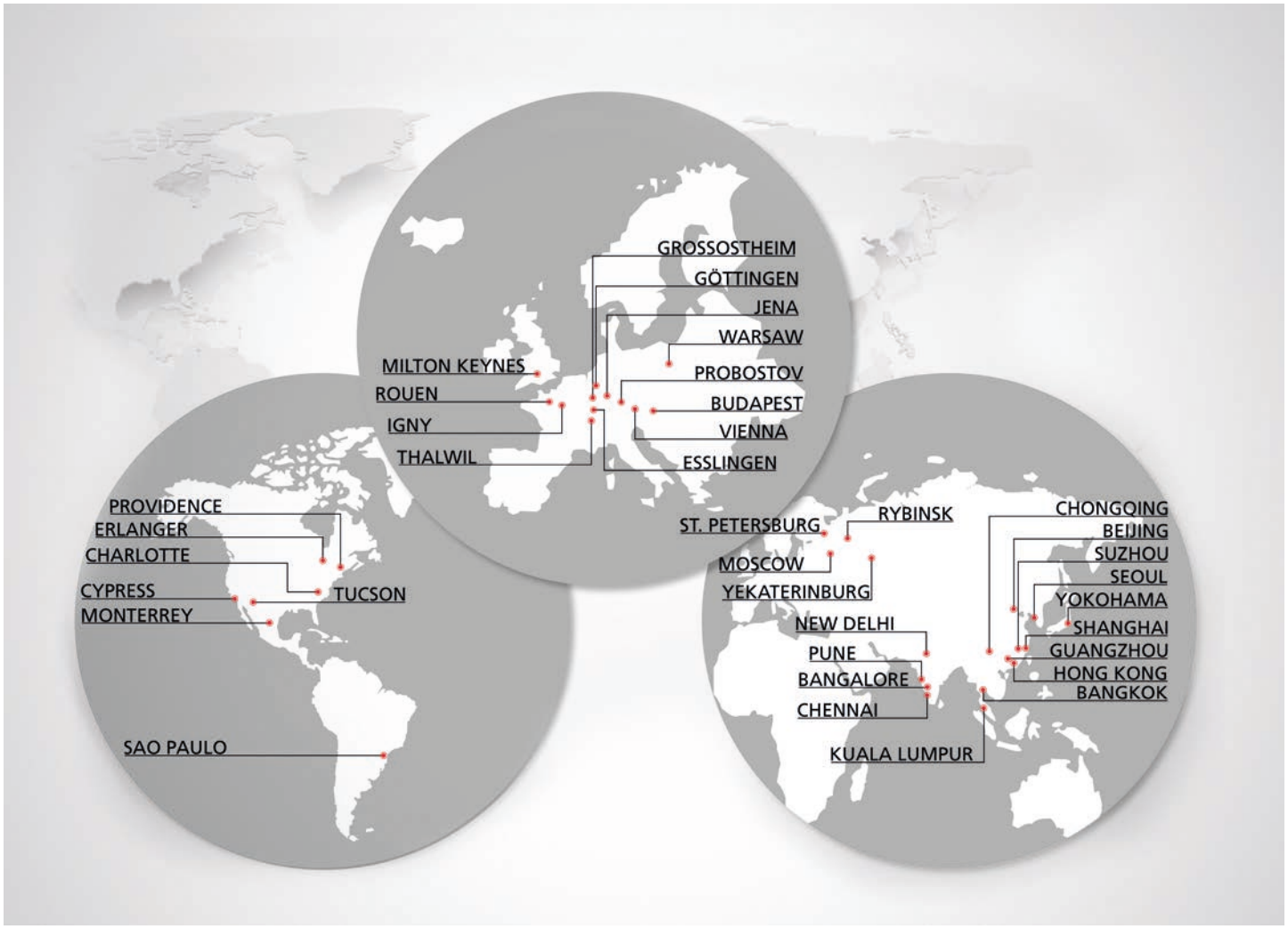
The new **Micromar 40 EWRi** digital micrometer micrometers have now also been equipped with the proven integrated wireless system. They also offer unique practical functions all over the world: 10 mm digits ensure fatigue-free reading of the measured values and the clear tolerance and warning limit display increases the process reliability in production and quality control. The new Micromar 40 EWRi electronic micrometer micrometers simplify and accelerate your quality assurance processes.



SMAHRT Solution

Customized **MarSolution measurement solutions** have long been an important product range of Mahr. Supporting the Industry 4.0 and a SMAHRT solution, networking, workpiece recognition by barcode reader or loading/unloading by robot is made possible. MWF Roland Friedrich also offers a complete range of services from project management and design / development of instruments to the turnkey handover, commissioning and training at the customer's site, to customer service with maintenance. The aim is to accompany businesses long term in the quality assurance of their workpieces, from the generation of an idea to the successful use of the equipment.

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Mahr GmbH, Germany
Carl-Mahr-Straße 1, 37073 Göttingen, Germany
Reutlinger Str. 48, 73728 Esslingen

Phone +49 551 7073-800, Fax +49 551 7073-888

info@mahr.de, www.mahr.com

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